Field reconstruction and curvature identifiability from spherical maps in a closed spatial geometry

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An angular map, a resolved spatial volume, and a statistical model of a field contain different information about a closed universe. We distinguish these inverse problems for scalar fields with the exact harmonic cutoff $0 \le n \le L$ on a round three-sphere. For q concentric spherical maps, we determine the null space explicitly as a product of q radial factors times the band of degree $L-q$. Its dimension is N_{L-q}, where $N_j = (j+1)(j+2)(2j+3)/6$ and $N_j = 0$ for $j < 0$. The resulting rank bound also applies to any q linear scalar channels equivariant under rotations about the observer. A resolved layer of angular half-width ε is injective, but its squared singular values separate into the powers ε^{2k} with multiplicities $(L-k+1)^2$; we give the exact leading coefficient of the weakest mode. We then prove that a single map has zero local Fisher information about its angular radius after unrestricted positive isotropic band powers are profiled out. In contrast, two shells with independently known radial distances determine the curvature radius from their highest angular sector, provided that sector is nonzero and the spectral cutoff is exact. We derive its noise information and a contamination bound. These are conditional results for specified observation operators, not a detection of spatial curvature or a derivation of cosmic acceleration.

I. INTRODUCTION

The geometry of a compact spatial slice does not specify which aspects of that geometry are identifiable from the observations available to one observer. In particular, access to a two-dimensional sky is not access to an open three-dimensional region. This distinction is essential when a geometrical cosmology is formulated through mode counting, analytic continuation, or a proposed correspondence between a global field and an observable domain.

Our earlier conditional reconstruction manuscripts [1, 2] distinguish an assumed round S^3 geometry from a measured topology and propose restriction and transfer operators as the next step. A companion working manuscript [3] treats finite-band scalar fields on three-dimensional caps. Here we address the different problem of observation on spherical surfaces, and distinguish recovery of a field at fixed geometry from identification of the geometry itself. No dark-energy, phase-evolution, or Fibonacci-scale postulate is needed for the results below.

The ingredients have substantial precedents. Polynomial factorization on latitudes is an established method of spherical interpolation [4]; Gaussian quadrature supplies exact finite integration rules [5]; spherical concentration operators quantify the stability of restricted observations [6]. Cosmological inference is known to depend on assumptions about the primordial spectrum [7], including in recent topology calculations [8]. Relative responses to a common random realization can evade a sample-variance limit under suitable assumptions [9].

Our contribution is a complete calculation within one explicitly defined S^3 observation model: the lost field subspace, its recovery by resolved radial information, an exact statistical degeneracy, and a conditional two-shell curvature reconstruction. The factorization and quadrature arguments are applications of established mathematics. We do not claim that their underlying principles, spectral discreteness, or cancellation of common realization amplitudes are newly discovered physical laws. The precise combined results are offered for independent mathematical and physical assessment; a search of related literature is not proof of historical priority.

II. GEOMETRY, FIELD SPACE, AND OBSERVATION MODEL

On a fixed round spatial slice of radius $R > 0$, use observer-centered coordinates

$$\mathrm{d}s^2 = R^2(\mathrm{d}\theta^2 + \sin^2\theta\,\mathrm{d}\Omega^2), \qquad 0 \le \theta \le \pi. \tag{1}$$

Write $\mathrm{d}\sigma = \mathrm{d}\Omega/(4\pi)$ and normalize the spatial measure to one:

$$\mathrm{d}\mu = \frac{\mathrm{d}V}{2\pi^2 R^3} = \frac{2}{\pi}\sin^2\theta\,\mathrm{d}\theta\,\mathrm{d}\sigma. \tag{2}$$

Let $Y_{\ell m}$ be orthonormal in $L^2(S^2, \mathrm{d}\sigma)$; thus $Y_{00} = 1$. Using real spherical harmonics avoids reality constraints on coefficients. The complex convention gives the same dimensions and spectra.

Let $\mathcal{H}_n$ be the scalar eigenspace of $-\Delta$ with eigenvalue $n(n+2)/R^2$. Its dimension is $(n+1)^2$. Our field space is the *complete* band

$$\mathcal{H}_{\le L} = \bigoplus_{n=0}^{L} \mathcal{H}_n, \tag{3}$$

$$N_L = \sum_{n=0}^{L} (n+1)^2 = \frac{(L+1)(L+2)(2L+3)}{6}. \tag{4}$$

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Set $\mathcal{H}_{\le j} = \{0\}$ and $N_j = 0$ for $j < 0$. The cutoff is an assumption about the field, not a consequence of compactness. Compactness alone leaves infinitely many eigenmodes.

An orthonormal basis is

$$Q_{n\ell m}(\theta,\Omega) = r_{n\ell}(\theta) Y_{\ell m}(\Omega), \tag{5}$$

$$r_{n\ell}(\theta) = a_{n\ell} (\sin\theta)^{\ell} C_{n-\ell}^{\ell+1}(\cos\theta), \tag{6}$$

$$a_{n\ell}^2 = \frac{2^{2\ell}(n-\ell)!(n+1)(\ell!)^2}{(n+\ell+1)!}, \quad 0 \le \ell \le n \le L. \tag{7}$$

The normalization follows by Gegenbauer orthogonality with weight $(1-t^2)^{\ell+1/2}$ [10]. In particular, $r_{n0} = C_n^1$ and $r_{LL} = a_{LL} \sin^L\theta$.

Since Gegenbauer polynomials of degrees zero through $L-\ell$ form a polynomial basis, every field has the unique representation

$$u = \sum_{\ell=0}^{L} \sum_{m=-\ell}^{\ell} (\sin\theta)^{\ell} p_{\ell m}(\cos\theta) Y_{\ell m}, \qquad \deg p_{\ell m} \le L-\ell. \tag{8}$$

This elementary separation is the basis of the proofs.

All surface maps below are scalar values of the *same* field on one spatial slice, with complete angular coverage and a known common center. Their norm uses $\mathrm{d}\sigma$, not physical surface area. Real cosmological maps generally lie on a past light cone and involve source evolution, line-of-sight integrals, lensing, and selection effects. Those operations must be included in a forward model before an observational application.

III. AN EXACT CEILING ON INFORMATION IN ANGULAR CHANNELS

Let T be a linear observation operator with q scalar sky maps as its output. Suppose it commutes with the $SO(3)$ rotations fixing the observer. In each angular sector, the field contains $L-\ell+1$ independent radial coefficients, whereas the output contains at most q.

Theorem 1 (Angular-channel rank bound)**.** *For an operator with these assumptions,*

$$\operatorname{rank} T \le \sum_{\ell=0}^{L} (2\ell+1) \min\{q, L-\ell+1\} \tag{9}$$

$$= N_L - N_{L-q}. \tag{10}$$

Consequently at least N_{L-q} field directions are unidentifiable.

Proof. The restriction of the field representation to rotations about the observer is

$$\mathcal{H}_{\le L} \cong \bigoplus_{\ell=0}^{L} \left(\mathbb{R}^{L-\ell+1} \otimes V_\ell\right), \qquad \dim V_\ell = 2\ell+1. \tag{11}$$

An equivariant linear map acts in this sector by a radial matrix $T_\ell : \mathbb{R}^{L-\ell+1} \to \mathbb{R}^q$, tensor the identity on V_ℓ. The rank of this matrix is at most the smaller dimension. For $q \le L$, the deficit is $\sum_{\ell=0}^{L-q}(2\ell+1)(L-q-\ell+1) = N_{L-q}$. For $q \ge L+1$ the deficit is zero. □

This statement includes finitely many radially averaged maps and any known rotationally invariant transfer functions. Widening a radial window does not increase the number of output channels. Increasing the angular pixel count beyond that required to represent the band cannot remove this radial rank deficit. A sky mask or an additional deterministic compression of such outputs cannot increase their rank.

A. Concentric shells attain the ceiling

Choose distinct $\chi_j \in (0,\pi)$, $1 \le j \le q$, and define

$$B_q u = \{u(\chi_j,\cdot)\}_{j=1}^{q}, \qquad P_q(t) = \prod_{j=1}^{q} (t - \cos\chi_j). \tag{12}$$

Arbitrary fixed positive weights may be used for the output maps; the kernel does not depend on them.

Theorem 2 (Complete shell null space)**.** *The observation kernel is exactly*

$$\boxed{\ker B_q = P_q(\cos\theta)\, \mathcal{H}_{\le L-q}.} \tag{13}$$

Its dimension is N_{L-q} and its rank is $N_L - N_{L-q}$. Thus $L+1$ distinct nonpolar shells are necessary and sufficient to recover every field in the complete band.

Proof. Project a zero shell map onto $Y_{\ell m}$. Since $\sin\chi_j \neq 0$, Eq. (8) gives $p_{\ell m}(\cos\chi_j) = 0$ for all j. Distinct roots imply that $p_{\ell m}$ is divisible by P_q, or is zero when its degree is less than q. The quotient has degree at most $L-q-\ell$, so the collection of quotients is precisely the polynomial representation of $\mathcal{H}_{\le L-q}$. Conversely every product in Eq. (13) belongs to $\mathcal{H}_{\le L}$ and vanishes on all the shells. Multiplication by $P_q(\cos\theta)$ is injective on the quotient space: a product zero everywhere has its second factor zero on the open set where $P_q \neq 0$, hence zero as a finite harmonic sum. The dimension and rank follow. □

For a single shell, the rank is only $(L+1)^2$, compared with $N_L \sim L^3/3$ for the spatial field. The constant mode is included in these counts. Deleting a measured monopole or dipole gives a different operator with additional missing information.

The necessity of $L+1$ shells is stronger than a comparison of total input and output dimensions: the monopole sector alone contains $L+1$ radial degrees of freedom. Angular modes cannot replace those measurements when the operator preserves observer-centered rotations.

Corollary 3 (Normal-derivative data)**.** *If at shell j one measures the derivatives of orders $0,\dots,s_j-1$ in the normal direction and $q=\sum_j s_j$, then*

$$\ker B_{\mathrm{jet}} = \prod_j (\cos\theta - \cos\chi_j)^{s_j}\,\mathcal{H}_{\le L-q}. \tag{14}$$

Proof. At a nonpolar radius $\theta \mapsto \cos\theta$ is a local coordinate change with nonzero derivative. The factor $\sin^\ell\theta$ is smooth and nonzero. Multiplication by that factor and the coordinate change preserve the order of vanishing. A zero normal jet therefore imposes the indicated root multiplicity on each $p_{\ell m}$. Apply polynomial division as before. The physical normal derivative $R^{-1}\partial_\theta$ changes nonzero scaling factors, not the kernel. □

B. An exact, optimally conditioned shell design

Proposition 4 (Gaussian shell isometry)**.** *Take $q=L+1$ shells with*

$$\chi_j = \frac{j\pi}{L+2}, \qquad w_j = \frac{2}{L+2}\sin^2\chi_j, \quad j=1,\dots,L+1. \tag{15}$$

Then for every $u\in\mathcal{H}_{\le L}$,

$$\sum_j w_j \int_{S^2} |u(\chi_j,\Omega)|^2 \mathrm{d}\sigma = \|u\|^2_{L^2(\mu)}\,. \tag{16}$$

Every singular value of this weighted observation operator is one.

Proof. Angular orthogonality reduces the integrand to $\sum_{\ell m}(1-t^2)^\ell |p_{\ell m}(t)|^2$, a real polynomial of degree at most $2L$. Equation (2) induces the normalized weight $(2/\pi)\sqrt{1-t^2}\,\mathrm{d}t$. The nodes and weights in Eq. (15) are its $(L+1)$-point Gauss–Chebyshev rule of the second kind, exact through degree $2L+1$ [5]. This proves the norm identity. Polarization gives the inner-product identity. □

This is an application of Gaussian quadrature, not a new quadrature rule. Its significance here is that the minimum shell count and perfect conditioning can be attained simultaneously if the full radial range is available and the geometry is known. These shells are not claimed to be accessible cosmological survey surfaces.

IV. A RESOLVED THIN LAYER: UNIQUENESS AND ITS COST

Fix $\chi_0\in(0,\pi)$ and let $0<\varepsilon<\min\{\chi_0,\pi-\chi_0\}$. In this section the data resolve the radial coordinate throughout the layer, rather than averaging it into one map. Let $w\ge 0$ be an integrable weight on $[-1,1]$, with $\int w\mathrm{d}s = 1$, positive almost everywhere on some interval. Define

$$\|A_\varepsilon u\|^2 = \int_{-1}^{1} w(s)\int_{S^2} |u(\chi_0+\varepsilon s,\Omega)|^2 \mathrm{d}\sigma\,\mathrm{d}s. \tag{17}$$

Use the full spatial norm (2) on the input. All following asymptotics hold at fixed L, χ_0, and w as $\varepsilon\to 0^+$. They are not uniform large-bandwidth statements.

Theorem 5 (Complete thin-layer spectrum)**.** *The operator A_ε is injective. Its squared singular values within angular sector ℓ can be ordered as*

$$\lambda_{\ell k}(\varepsilon) = \Theta(\varepsilon^{2k}), \quad k=0,\dots,L-\ell, \tag{18}$$

each repeated $2\ell+1$ times. In the complete spectrum, the scaling power ε^{2k} has multiplicity $(L-k+1)^2$. For $L\ge 1$, the unique weakest branch has $\ell=0$, $k=L$ and

$$\lambda_{\min}(\varepsilon) \sim \frac{\sin^{2L}\chi_0\, h_L(w)}{J_L(\cos\chi_0)}\,\varepsilon^{2L}, \tag{19}$$

where

$$h_L(w) = \min_{b_0,\dots,b_{L-1}} \int_{-1}^{1} w(s)\left| s^L + \sum_{j<L} b_j s^j\right|^2 \mathrm{d}s, \tag{20}$$

$$J_L(c) = \frac{2}{\pi}\int_{-1}^{1} (t-c)^{2L}\sqrt{1-t^2}\,\mathrm{d}t. \tag{21}$$

For $L=0$, $\lambda_{\min}=1$ and the same coefficient formula applies.

Proof. Use in sector ℓ the basis $\sin^\ell\theta(\cos\theta-c)^i$, $0\le i\le M=L-\ell$, where $c=\cos\chi_0$. Denote its full spatial Gram matrix by F_ℓ. It is fixed and positive definite. Its observation Gram matrix obeys

$$G_\ell(\varepsilon) = D_\varepsilon M_\ell(\varepsilon) D_\varepsilon, \quad D_\varepsilon = \mathrm{diag}(1,\varepsilon,\dots,\varepsilon^M). \tag{22}$$

Indeed, uniform Taylor expansion on $s\in[-1,1]$ gives

$$\begin{aligned} \big[M_\ell(0)\big]_{ij} &= \sin^{2\ell}\chi_0(-\sin\chi_0)^{i+j}\mu_{i+j}, \\ \mu_r &= \int_{-1}^{1} s^r w(s)\mathrm{d}s. \end{aligned} \tag{23}$$

The limiting moment matrix is positive definite because a nonzero polynomial cannot vanish on an interval. Continuity therefore gives fixed positive upper and lower matrix bounds on $M_\ell(\varepsilon)$ for sufficiently small ε. Comparing the Rayleigh quotient $a^*G_\ell a/(a^*F_\ell a)$ with $a^*D_\varepsilon^2 a/(a^*a)$ and applying the finite-dimensional min–max principle proves Eq. (18). At each fixed admissible ε, the same polynomial argument proves positive definiteness and injectivity. Summing $2\ell+1$ over $0\le\ell\le L-k$ gives $(L-k+1)^2$.

For the weakest branch only $\ell=0$ contributes at exponent $2L$. The reciprocal eigenvalues are those of $F_0^{1/2}G_0^{-1}F_0^{1/2}$. From Eq. (22),

$$\varepsilon^{2L}G_0^{-1} \longrightarrow [M_0(0)^{-1}]_{LL} e_L e_L^*. \tag{24}$$

Its nonzero eigenvalue after multiplication by $F_0^{1/2}$ on both sides is $[M_0(0)^{-1}]_{LL}[F_0]_{LL}$. The Schur complement formula for the monic orthogonal polynomial gives $([\mu_{i+j}]^{-1})_{LL} = 1/h_L(w)$. Equation (23) then yields $[M_0(0)^{-1}]_{LL} = \sin^{-2L}\chi_0/h_L(w)$, while $[F_0]_{LL}=J_L(c)$. Taking the reciprocal proves Eq. (19). □

For the uniform weight $w = 1/2$, the leading coefficient of the Legendre polynomial gives

$$h_L(1/2) = \frac{2^{2L}}{(2L+1)\binom{2L}{L}^2}. \tag{25}$$

The spatial moment can be evaluated without quadrature:

$$J_L(c) = \sum_{r=0}^{L} \binom{2L}{2r} c^{2L-2r} \frac{\binom{2r}{r}}{4^r(r+1)}. \tag{26}$$

At the equator and $L = 1$, for example, $\lambda_{\min} \sim 4\varepsilon^2/3$. The exponents describe multiplicities of asymptotic orders, not exact degeneracies between different ℓ.

The inverse norm grows as

$$\left\|A_\varepsilon^+\right\| \sim \left[\frac{J_L(\cos\chi_0)}{\sin^{2L}\chi_0 h_L(w)}\right]^{1/2} \varepsilon^{-L}. \tag{27}$$

Here the pseudoinverse acts on the finite-dimensional signal range. Thus the infinite precision needed in the zero-width limit is quantified, even though any positive resolved thickness gives uniqueness. In the limit, exactly $(L+1)^2$ singular-value branches remain of order one, matching the rank of one shell.

The normalization in Eq. (17) matters. For the actual unrenormalized spatial norm restricted to a uniform thin layer,

$$\int_{\chi_0-\varepsilon}^{\chi_0+\varepsilon} \frac{2}{\pi} \sin^2\theta \mathrm{d}\theta \int |u|^2 \mathrm{d}\sigma, \tag{28}$$

every exponent increases by one. The weakest coefficient is multiplied by $4\sin^2\chi_0/\pi$ relative to Eq. (19) with $w = 1/2$. A physically specified noise model must determine which norm describes the experiment.

A. Noise and indistinguishability

For finitely many real signal coefficients with Gaussian additive noise $\mathcal{N}$ of positive-definite covariance, the Fisher matrix is $F = T^*\mathcal{N}^{-1}T$. Hence $\ker F = \ker T$. If $v \in \ker T$ has unit norm, the fields $u+av$ and $u-av$ have identical data distributions. For any estimator, their maximum mean squared error is at least a^2: the average of their two risks equals the risk about their midpoint plus a^2. No assertion about exact reconstruction can overcome this null space without additional information.

For resolved-layer data, project the observations onto an orthonormal basis of the finite-dimensional signal range and assume coefficient noise of variance σ^2. This specifies the otherwise ambiguous notion of white continuum noise. The Cramér–Rao variance in the weakest direction is $\sigma^2/\lambda_{\min}$ and therefore grows as ε^{-2L}. A different noise covariance uniformly bounded above and below, independently of ε, preserves the powers but changes the coefficients.

V. A SINGLE MAP CANNOT IDENTIFY ITS RADIUS WITH FREE BAND POWERS

Now let the input be a zero-mean $SO(4)$-isotropic real Gaussian field. In the orthonormal basis (7), its coefficients satisfy

$$\mathbb{E}[x_{n\ell m}x_{n'\ell'm'}] = P_n\delta_{nn'}\delta_{\ell\ell'}\delta_{mm'}, \qquad P_n > 0. \tag{29}$$

There is one unrestricted band power per spatial degree n. For a shell at angular radius χ, the sky covariance is diagonal:

$$C_\ell(\chi, P) = \sum_{n=\ell}^{L} M_{\ell n}(\chi) P_n, \tag{30}$$

$$M_{\ell n}(\chi) = r_{n\ell}(\chi)^2, \quad n \geq \ell, \tag{31}$$

and $M_{\ell n} = 0$ for $n < \ell$.

Theorem 6 (Exact local statistical degeneracy)**.** *Fix $\chi_0 \in (0, \pi)$ and any strictly positive vector $P^{(0)}$. There is an open interval around χ_0 on which*

$$P(\chi) = M(\chi)^{-1}M(\chi_0)P^{(0)} > 0 \tag{32}$$

and the entire Gaussian sky distribution is unchanged. The Fisher information for χ after profiling over all P_n is identically zero at every such interior parameter point. If an independently fixed radial distance d defines $\chi = d/R$, the same local nonidentifiability applies to R.

Proof. The matrix M is upper triangular with diagonal

$$M_{\ell\ell} = a_{\ell\ell}^2 \sin^{2\ell}\chi > 0. \tag{33}$$

It is therefore invertible and smooth for all $0 < \chi < \pi$. Equation (32) preserves $C = MP$ exactly. Positivity is preserved in a sufficiently small interval by continuity and the finite number of strictly positive components. A zero-mean Gaussian field is determined by its covariance, so this equality concerns the full map likelihood, not just a single realization or an approximate fit.

For the Fisher statement, the derivative $M'(\chi)P$ lies in the span of the band-power derivative columns of M. Thus the score for χ is a linear combination of the nuisance-parameter scores. Its residual after projection onto their span vanishes. Equivalently, $P' = -M^{-1}M'P$ is a tangent vector of constant likelihood in Eq. (32). The change of variable $\chi = d/R$ preserves the degeneracy. □

This result is local in the interior of the positive-power cone. It does not assert that every positive sky spectrum is realizable at every radius. At a boundary where some $P_n = 0$, positivity can constrain a variation. A restricted primordial spectrum, dynamics, cross-correlated channels, or external distance information can change the problem. Known additive noise independent of χ does not remove the equality of distributions. A parameter-independent mask or other deterministic processing also preserves it.

There is a further exact ambiguity even with fixed band powers: $M(\pi-\chi)=M(\chi)$. This follows from the parity of Gegenbauer polynomials. An angular-radius prior restricting the domain can remove this antipodal pair, but not the local free-spectrum degeneracy above.

For $L=1$, the model is particularly transparent:

$$C_0 = P_0 + 4\cos^2\chi\, P_1, \qquad C_1 = \frac{4}{3}\sin^2\chi\, P_1. \tag{34}$$

Changing χ can be compensated by $P_1 = 3C_1/(4\sin^2\chi)$ and $P_0 = C_0 - 3C_1\cot^2\chi$, while these remain positive.

The theorem concerns a pure shell-value model on a fixed complete finite band. It is not a no-go theorem for curvature measurements using the actual cosmic microwave background (CMB), whose acoustic transfer functions, polarization, and cosmological priors supply other structure. It does show why a spectral gap or a projected angular spectrum alone cannot establish curvature without specifying the allowed source spectra.

VI. TWO SHELLS CAN IDENTIFY CURVATURE WITHOUT RECONSTRUCTING THE FIELD

Suppose two geodesic radii $0 < d_1 < d_2$ on the same slice are independently calibrated in common length units. Let

$$\kappa = \frac{d_2}{d_1} > 1, \qquad \chi = \frac{d_1}{R}, \qquad R > \frac{d_2}{\pi}. \tag{35}$$

The branch condition places both shells before the antipode. It is necessary for the geometric interpretation used here.

Theorem 7 (Conditional two-shell curvature reconstruction)**.** *Assume the exact cutoff $L \geq 1$ and a nonzero projection of the field onto the angular sector $\ell = L$ about the common observer. Let b_j be the real vector of its $2L+1$ measured coefficients on shell j. Then*

$$b_2 = g(R)\, b_1, \qquad g(R) = \left[\frac{\sin(d_2/R)}{\sin(d_1/R)}\right]^L. \tag{36}$$

The measured ratio $g = \langle b_1, b_2\rangle / \|b_1\|^2$ determines exactly one radius in Eq. (35), whenever $0 < g^{1/L} < \kappa$. This determination requires neither the unknown realization coefficients nor the positive band power P_L.

Proof. Only $n = L$ contributes to angular degree $\ell = L$ in the complete band. Equation (7) therefore gives $b_j = a_{LL}\sin^L(d_j/R)\, x_L$, where x_L is the common nonzero vector. This proves Eq. (36) and the amplitude cancellation.

It remains to prove uniqueness, not just write an inversion equation. Define $f_\kappa(x) = \sin(\kappa x)/\sin x$ on $0 < x < \pi/\kappa$. Its logarithmic derivative is

$$\frac{f_\kappa'(x)}{f_\kappa(x)} = \kappa\cot(\kappa x) - \cot x < 0. \tag{37}$$

To establish the inequality, put $h(x) = x\cot x$. Then $h'(x) = [\sin x\cos x - x]/\sin^2 x < 0$ on $(0,\pi)$: the numerator is zero at zero and has derivative $-2\sin^2 x$. Since $\kappa x > x$, $h(\kappa x) < h(x)$, proving Eq. (37). Finally, $f_\kappa(0^+) = \kappa$ and $f_\kappa((\pi/\kappa)^-) = 0$. Every admissible ratio therefore has exactly one x, and $R = d_1/x$. □

In the isotropic Gaussian model with $P_L > 0$, the required vector is nonzero with probability one. Deterministic fields can have a zero projection in that sector even if they contain other degree-L modes; the theorem explicitly excludes this case.

For $\kappa = 2$, there is a closed form:

$$\boxed{R = \frac{d_1}{\arccos\left(g^{1/L}/2\right)}}, \qquad 0 < g^{1/L} < 2. \tag{38}$$

The same ratio can be written from exact ensemble covariances as $g = C_L^{12}/C_L^{11}$, and $(C_L^{12})^2 = C_L^{11}C_L^{22}$. In the ideal noiseless model it already holds realization by realization, so it does not require an ensemble estimate. The cancellation uses the same principle of relative response to a common realization that appears in multitracer inference [9]; the sine-ratio inversion is specific to the present highest-sector shell model.

For $L \geq 2$, two shells still leave N_{L-2} field directions undetermined at the recovered radius. Thus identification of one geometrical parameter is compatible with failure of full field reconstruction. For $L = 1$, two shells also suffice for the field once the geometry has been determined.

A. Noise information and the nearly flat limit

Write the two observed top-sector vectors as

$$y_1 = a + z_1, \qquad y_2 = g(R)a + z_2, \qquad z_j \sim \mathcal{N}(0, \sigma_j^2 I), \tag{39}$$

with independent noises and an unrestricted deterministic nuisance vector $a \neq 0$. Taking the Schur complement over a gives the information

$$I_g = \frac{\|a\|^2}{\sigma_2^2 + g^2\sigma_1^2}, \qquad I_R = \frac{[g'(R)]^2\,\|a\|^2}{\sigma_2^2 + g^2\sigma_1^2}. \tag{40}$$

For completeness, before profiling the blocks are $I_{gg} = \|a\|^2/\sigma_2^2$, $I_{ga} = ga^T/\sigma_2^2$, and $I_{aa} = (\sigma_1^{-2} + g^2\sigma_2^{-2})I$. These produce Eq. (40) directly. This is a conditional noise calculation, not a forecast for a real survey or a claim about the bias of the simple ratio estimator in noise.

Let $K = R^{-2}$ denote the positive sectional curvature, distinct from the cosmological parameter Ω_K. For fixed calibrated distances,

$$\ln g = L\ln\kappa - \frac{L(d_2^2 - d_1^2)}{6}K + O(K^2). \tag{41}$$

Thus the ratio tends to its flat-space value κ^L, and $\partial_R \ln g = L(d_2^2 - d_1^2)/(3R^3) + O(R^{-5})$. At fixed nonzero reference signal a and noise levels, I_R decreases as R^{-6}. The information for K can remain finite at $K = 0$ under this normalization, but finite precision cannot imply a relative precision on an arbitrarily small curvature. If the physical source amplitude changes with R, it must also be included in the forecast.

B. What breaks the reconstruction

The exact upper spectral edge is central. If modes with $n > L$ are present, they contribute to $\ell = L$ with different radial factors and the common-vector relation need not hold. Applying an angular low-pass filter does not impose the required three-dimensional spectral cutoff. Let the two noiseless maps be

$$y_1 = a + e_1, \qquad y_2 = ga + e_2, \tag{42}$$

where e_j includes any such contamination. Define $\widehat{g} = \langle y_1, y_2\rangle \,/\, \|y_1\|^2$. Then the exact identity

$$\widehat{g} - g = \frac{\langle y_1, e_2 - g e_1\rangle}{\|y_1\|^2} \tag{43}$$

implies, if $\|e_j\| \le \eta_j \|a\|$ and $\eta_1 < 1$,

$$|\widehat{g} - g| \le \frac{\eta_2 + g\eta_1}{1 - \eta_1}. \tag{44}$$

This bound requires externally justified contamination bounds; it cannot be used to infer them from the two maps alone. On a compact interval of admissible R, monotonicity converts a ratio error into a radius error using the minimum of $g'(R) > 0$ on that interval.

Unknown relative calibration is another exact obstruction: if the maps have an unknown gain ratio b, the observable ratio is $b\,g(R)$ and one ratio cannot determine both parameters. Source evolution between two light-cone shells can produce the same problem. Finally, d_1 and d_2 must not be obtained by assuming the curvature that is being tested. Known κ alone gives χ; one independently known absolute distance is needed to turn it into a dimensionful radius.

VII. RELATION TO COSMOLOGICAL RECONSTRUCTION

Theorems 2, 5, 6, and 7 answer different questions. At fixed geometry, finitely many maps have an exact field null space. A resolved layer removes that null space with a calculable loss of stability as its width decreases. With unknown geometry, a freely adjustable positive spectrum can erase single-map sensitivity to radius exactly. A second calibrated response to the same highest-sector realization can restore that sensitivity without restoring all field coefficients.

No assumed volume fraction is an information fraction in these statements. For example, the cap fraction

$$f(\chi) = \frac{\chi - \sin\chi\cos\chi}{\pi} \tag{45}$$

does not determine the rank or the noise-conditioned singular values of a sky-transfer operator. Nor does analytic continuation from an open volume imply continuation from one embedded two-sphere: the explicit kernel (13) is a counterexample.

For a physical application, a scalar channel at fixed angular degree has a transfer matrix of the form

$$y_{\alpha,\ell m} = \sum_{n=\ell}^{L} T_{\alpha\ell n}(R, \vartheta) x_{n\ell m} + z_{\alpha,\ell m}, \tag{46}$$

where ϑ includes evolution, calibration, and source parameters. At fixed parameters, Theorem 1 supplies an exact upper bound for q such scalar channels. The actual rank may be lower. For parameter identification, one must additionally differentiate the likelihood and remove nuisance directions, as in Theorem 6. An invertible field map at fixed R does not by itself prove that R is identifiable.

The CMB, polarization, and galaxy observations do not have the unmodified shell transfer used in Theorem 7. Their source fields, removed low multipoles, and redshift-dependent transfer functions must be treated explicitly. In particular, neither a sharp cutoff nor its location has been established by the supplied manuscripts. The present reconstruction is therefore a conditional benchmark for developing such an experiment, not an observational determination of the Universe's radius, topology, dark-energy content, or initial state.

VIII. VERIFICATION AND RESEARCH METHODS

The analytic arguments are finite-dimensional proofs. Numerical checks are independent consistency tests of formulas and normalizations, not substitutes for those proofs. Appendix A contains the full Python program used here, with no input files and no third-party packages. It uses exact rational row reduction for shell and confluent ranks, and 160-digit decimal arithmetic for moment matrices, normalization, singular values, and covariance degeneracies.

The resolved-layer Gram matrices are evaluated by finite trigonometric moment sums. For uniform radial weight,

$$\begin{aligned} \langle \cos^n\theta\rangle_{\text{layer}} &= 2^{-n}\sum_{j=0}^{n}\binom{n}{j}\cos[(n-2j)\chi_0] \\ &\quad\times \operatorname{sinc}[(n-2j)\varepsilon], \\ \langle \cos^{2r}\theta\rangle_{\mu} &= \frac{\binom{2r}{r}}{4^r(r+1)}, \\ \langle \cos^{2r+1}\theta\rangle_{\mu} &= 0. \end{aligned} \tag{47}$$

TABLE I. Synthetic thin-layer check. c_L is the coefficient in Eq. (19); the last column is $\lambda_{\min}(0.01)/(c_L\,0.01^{2L})$.

L	c_L	Finite-width ratio
0	1	1.000000000
1	0.58091171	0.999970550
2	0.11781344	1.000037077
3	0.018139078	1.000061991
4	0.0024629588	1.000082387
5	0.00031074507	1.000101175
6	3.7342137e-05	1.000119207

with $\mathrm{sinc}(0) = 1$. These expressions avoid a quadrature resolution error as the layer narrows. Full-norm Cholesky whitening and symmetric Jacobi diagonalization then give the squared singular values. No astronomical observations or likelihood chains are used.

The executed program completed 774 assertions. Exact shell ranks were checked for $0 \le L \le 8$ and $1 \le q \le L+2$, and confluent ranks for $0 \le L \le 6$. Gaussian-shell moments through their required degree agreed with independent semicircle moments to better than 10^{-125}. The local positive-spectrum degeneracy was checked for $1 \le L \le 7$, at $\chi = 0.9$ and 0.90001 radians, with covariance residuals below 10^{-125}.

For the thin-layer tests, $\chi_0 = 1.1$ radians and $w = 1/2$. Slopes from widths 0.04 and 0.02 agreed with all predicted powers through $L = 6$ within 0.012. Table I reports the weakest coefficient and its independently computed finite-width ratio. Two-shell inversions with $d_1 = 3$, $d_2 = 6$, and $R = 10$ in common length units recovered R to relative error below 10^{-100} for $L = 1, 2, 4, 7$. The Fisher Schur complement was also checked with exact fractions. These high-precision arithmetic residuals are not error bars on a physical measurement.

A. AI-assisted research disclosure

OpenAI ChatGPT, accessed through Codex Work Mode on September 9, 2026, was used substantively for literature synthesis, derivation assistance, proof checking, and generation and execution of the verification program. The backend model version was not exposed in the session metadata. Boris Batenin supplied the source manuscripts, defined the research direction, and requested explicit proofs and executable checks. Verification of the AI-assisted outputs in this workflow consists of the analytic derivations and reproducible numerical consistency tests described in this section; the packaged program was rerun and completed the 774 assertions reported above. The named authors retain responsibility for the scientific content and the final submitted text.

IX. DISCUSSION

The sharply stated result is an observability distinction. A finite set of angular channels can have a provable field null space while retaining information about a geometric parameter. Conversely, a map with infinitely precise angular coefficients can contain no local information about its radius when its source spectrum is free. These conclusions follow from the operator and nuisance space, not from spatial compactness alone.

The strongest route to a physical test is to replace the ideal shell responses by a calibrated light-cone transfer calculation, assess its nuisance-projected Fisher information, and bound leakage from modes outside any effective band. The exact null space, thin-layer coefficient, Gaussian-shell isometry, and two-shell monotonicity provide separate checks on that calculation. Failure of a proposed transfer model to pass one of these checks can reveal an information assumption before it is mistaken for a cosmological prediction.

ACKNOWLEDGMENTS

OpenAI ChatGPT also assisted with English drafting and LaTeX preparation. Its substantive research use is disclosed in Sec. VIII. No AI system is listed as an author.

DATA AVAILABILITY

No observational or experimental data were generated or analyzed in this study. The synthetic calculations supporting the reported consistency checks are reproducible from the complete Python verification program in Appendix A. The program requires only the Python standard library, prints the numerical results, and writes a local JSON report.

Appendix A: Complete verification program

Copy the following listing into `shell_verification.py` and run `python3 shell_verification.py`. Assertion failures stop the program. The fixed tolerances test only the finite cases stated in Sec. VIII; the all-L claims are proved in the text.

```
"""Synthetic checks for the spherical-shell manuscript;
no observational input."""
from decimal import Decimal as D, localcontext
from fractions import Fraction as Q
from math import comb, factorial
from pathlib import Path
import json

PI = D('3.14159265358979323846264338327950288419716939937510'
       '58209749445923078164062862089986280348253421170679'
       '82148086513282306647093844609550582231725359408128'
       '48111745028410270193 85')
```

```
def trig(x, sine=False):
    term = total = x if sine else D(1)
    for r in range(1, 400):
        j = 2*r + int(sine)
        term *= -x*x / D(j*(j-1))
        total += term
        if abs(term) < D('1e-155'):
            return total
    raise ArithmeticError('trigonometric series')


def mul(a, b):
    return [[sum(x*y for x, y in zip(row, col))
             for col in zip(*b)] for row in a]


def trans(a):
    return [list(c) for c in zip(*a)]


def chol(a):
    n = len(a)
    b = [[D(0)]*n for _ in range(n)]
    for i in range(n):
        for j in range(i+1):
            v = a[i][j] - sum(b[i][k]*b[j][k] for k in range(j))
            b[i][j] = v.sqrt() if i == j else v/b[j][j]
    return b


def invlower(a):
    n = len(a)
    b = [[D(0)]*n for _ in range(n)]
    for j in range(n):
        for i in range(n):
            b[i][j] = (D(i == j) - sum(a[i][k]*b[k][j]
                       for k in range(i)))/a[i][i]
    return b


def eig(a):
    a = [r[:] for r in a]
    n = len(a)
    if n == 1:
        return [a[0][0]]
    for _ in range(400*n*n):
        p, q = max(((i, j) for i in range(n) for j in range(i+1, n
     )),
                   key=lambda ij: abs(a[ij[0]][ij[1]]))
        off = a[p][q]
        if abs(off) < D('1e-125'):
            return sorted((a[i][i] for i in range(n)), reverse=
     True)
        tau = (a[q][q]-a[p][p])/(2*off)
        t = D(1 if tau >= 0 else -1)/(abs(tau)+(1+tau*tau).sqrt())
        c = 1/(1+t*t).sqrt()
        s = t*c
        for k in range(n):
            if k not in (p, q):
                kp, kq = a[k][p], a[k][q]
                a[k][p] = a[p][k] = c*kp-s*kq
                a[k][q] = a[q][k] = s*kp+c*kq
        a[p][p] -= t*off
        a[q][q] += t*off
        a[p][q] = a[q][p] = D(0)
    raise ArithmeticError('Jacobi convergence')


def N(L):
    return (L+1)*(L+2)*(2*L+3)//6 if L >= 0 else 0


def semicircle(n):
    return (D(comb(n, n//2))/D(n//2+1)/D(2)**n
            if n % 2 == 0 else D(0))


def interval_moment(n, chi, eps):
    ans = D(0)
    for k in range(n+1):
        m = n-2*k
        sinc = trig(m*eps, True)/(m*eps) if m else D(1)
        ans += D(comb(n, k))*trig(m*chi)*sinc
    return ans/D(2)**n


def moment(ell, r, center, base):
    return sum(D(comb(r, j))*(-center)**(r-j)
               * D(comb(ell, a))*(-1)**a*base(j+2*a)
               for j in range(r+1) for a in range(ell+1))


def matrices(L, ell, chi, eps):
    c = trig(chi)
    m = L-ell+1
    full = [[moment(ell, i+j, c, semicircle) for j in range(m)]
            for i in range(m)]
    obs = [[moment(ell, i+j, c,
             lambda n: interval_moment(n, chi, eps)) for j in
     range(m)]
           for i in range(m)]
    return obs, full


def spectrum(L, ell, chi, eps):
    obs, full = matrices(L, ell, chi, eps)
    b = invlower(chol(full))
    return eig(mul(mul(b, obs), trans(b)))


def rational_rank(a):
    a = [list(map(Q, row)) for row in a]
    row = 0
    for col in range(len(a[0])):
        pivot = next((r for r in range(row, len(a)) if a[r][col]),
       None)
        if pivot is None:
            continue
        a[row], a[pivot] = a[pivot], a[row]
        v = a[row][col]
        a[row] = [x/v for x in a[row]]
        for r in range(len(a)):
            if r != row:
                v = a[r][col]
                a[r] = [x-v*y for x, y in zip(a[r], a[row])]
        row += 1
        if row == len(a):
            break
    return row


def gegen(k, lam, t):
    if k == 0:
        return D(1)
    old, cur = D(1), 2*lam*t
    for j in range(1, k):
        old, cur = cur, (2*(j+lam)*t*cur-(j+2*lam-1)*old)/(j+1)
    return cur


def radial(n, ell, chi):
    norm2 = (D(2)**(2*ell)*D(factorial(n-ell)*(n+1)*factorial(ell)
      **2)
             / D(factorial(n+ell+1)))
    return norm2.sqrt()*trig(chi, True)**ell*gegen(n-ell, ell+1,
      trig(chi))


def mixing(L, chi):
    return [[radial(n, ell, chi)**2 if n >= ell else D(0)
             for n in range(L+1)] for ell in range(L+1)]


def upper_solve(a, b):
    x = [D(0)]*len(b)
    for i in reversed(range(len(b))):
        x[i] = (b[i]-sum(a[i][j]*x[j] for j in range(i+1, len(b)))
      )/a[i][i]
    return x


def verify():
    checks = 0
    thin, rank_rows, curvature_rows = [], [], []
    max_isometry = D(0)
```

```
max_covariance_residual = D(0)
with localcontext() as ctx:
    ctx.prec = 160
    for L in range(9):
        for q in range(1, L+3):
            nodes = [Q(2*j-q+1, q+1) for j in range(q)]
            rank = sum((2*ell+1)*rational_rank(
                [[t**k for k in range(L-ell+1)] for t in nodes
])
                for ell in range(L+1))
            assert rank == N(L)-N(L-q)
            checks += 1
        rank_rows.append([L, N(L), N(L)-N(L-1), N(L)-N(L-2)])

        # Gaussian quadrature is checked against independent
        # semicircle moments.
        angles = [D(j)*PI/D(L+2) for j in range(1, L+2)]
        weights = [D(2)/D(L+2)*trig(a, True)**2 for a in
angles]
        for ell in range(L+1):
            for r in range(2*(L-ell)+1):
                observed = sum(w*trig(a, True)**(2*ell)*trig(a
)**r
                               for a, w in zip(angles, weights
))
                full = sum(D(comb(ell, k))*(-1)**k*semicircle(
r+2*k)
                           for k in range(ell+1))
                error = abs(observed-full)
                max_isometry = max(max_isometry, error)
                assert error < D('1e-125')
                checks += 1

    # Confluent ranks, including degrees lower than total jet
count.
    for L in range(7):
        for ell in range(L+1):
            rows = []
            for t, multiplicity in [(Q(-1, 3), 2), (Q(2, 5),
3)]:
                for derivative in range(multiplicity):
                    rows.append([Q(factorial(k), factorial(k-
derivative))
                                 * t**(k-derivative)
                                 if k >= derivative else Q(0)
                                 for k in range(L-ell+1)])
            assert rational_rank(rows) == min(5, L-ell+1)
            checks += 1

    for L in range(7):
        counts = {k: 0 for k in range(L+1)}
        largest_slope_error = D(0)
        for ell in range(L+1):
            first = spectrum(L, ell, D('1.1'), D('.04'))
            second = spectrum(L, ell, D('1.1'), D('.02'))
            for k, (v1, v2) in enumerate(zip(first, second)):
                slope = (v1/v2).ln()/D(2).ln()
                largest_slope_error = max(
                    largest_slope_error, abs(slope-2*k))
                assert abs(slope-2*k) < D('.08')
                counts[k] += 2*ell+1
                checks += 1
        assert counts == {k: (L-k+1)**2 for k in counts}
        checks += 1
        c = trig(D('1.1'))
        h = D(2)**(2*L)/D((2*L+1)*comb(2*L, L)**2)
        full_LL = moment(0, 2*L, c, semicircle)
        coefficient = trig(D('1.1'), True)**(2*L)*h/full_LL
        ratio = spectrum(L, 0, D('1.1'), D('.01'))[-1]/(
            coefficient*D('.01')**(2*L))
        assert abs(ratio-1) < D('.01')
        checks += 1
        thin.append({'L': L, 'coefficient': str(coefficient),
                     'ratio_at_0.01': str(ratio),
                     'max_slope_error': str(
largest_slope_error)})

    for L in range(1, 8):
        p0 = [D(1)/D(n+1)**2 for n in range(L+1)]
        m0, m1 = mixing(L, D('.9')), mixing(L, D('.90001'))
        cov = [sum(a*b for a, b in zip(row, p0)) for row in m0
]
        p1 = upper_solve(m1, cov)
        assert min(p1) > 0
        residual = max(abs(sum(a*b for a, b in zip(row, p1))-c
)
                       for row, c in zip(m1, cov))
        max_covariance_residual = max(max_covariance_residual,
 residual)
        assert residual < D('1e-125')
        checks += 1

    # Closed-form kappa=2 curvature reconstruction and
    # independent top-mode ratio.
    for L in (1, 2, 4, 7):
        R, d1, d2 = D(10), D(3), D(6)
        g = (trig(d2/R, True)/trig(d1/R, True))**L
        assert abs(g-radial(L, L, d2/R)/radial(L, L, d1/R)) <
D('1e-125')
        eta = g**(D(1)/D(L))
        # Invert cos by bisection; no trigonometric inverse
dependency.
        lo, hi = D(0), PI/2
        for _ in range(450):
            mid = (lo+hi)/2
            if 2*trig(mid) > eta:
                lo = mid
            else:
                hi = mid
        recovered = d1/((lo+hi)/2)
        assert abs(recovered/R-1) < D('1e-100')
        curvature_rows.append(
            {'L': L, 'g': str(g), 'R_recovered': str(recovered
)})
        checks += 1

    # Strict monotonicity, including the branch near the
antipode.
    for kap in (D('1.2'), D(2), D('3.7')):
        for j in range(1, 100):
            x = PI*D(j)/(100*kap)
            derivative = (kap*trig(kap*x)/trig(kap*x, True)
                          - trig(x)/trig(x, True))
            assert derivative < 0
            checks += 1

    # Exact fractional Fisher Schur complement for
deterministic
    # nuisance amplitudes.
    g, v1, v2, norm2 = Q(7, 5), Q(2), Q(3), Q(11)
    direct = norm2/v2-(g*g*norm2/v2**2)/(1/v1+g*g/v2)
    assert direct == norm2/(v2+g*g*v1)
    checks += 1

    report = {'status': 'PASS', 'checks': checks, '
decimal_digits': 160,
              'rank_table': rank_rows, 'thin_shell': thin,
              'max_isometry_residual': str(max_isometry),
              'max_covariance_residual': str(
max_covariance_residual),
              'two_shell_reconstruction': curvature_rows}
    Path('shell_verification.json').write_text(json.dumps(report,
     indent=2))
    print(json.dumps(report, indent=2))


if __name__ == '__main__':
    verify()
```

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